Runbook — Confluent Cloud for Apache Flink (CCAF)

End-to-end operational guide for running the `confluent-kafka-isotope` reports on **Confluent Cloud for Apache Flink (CCAF)**: provision → deploy reports → drive traffic → observe → teardown. Unlike the Confluent Platform + Flink on minikube, this is **Terraform-driven** — no local cluster — and everything runs in a fresh Confluent Cloud environment under terraform/.

> This is the **managed CCAF** path. The self-managed **Confluent Platform + Flink on minikube** path is in docs/runbook-local.md. Both runtimes run the same seven reports from the same PTF JAR — see root README §3.3 for the format-by-runtime split.

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1.0 Prerequisites

- Terraform `>= 1.17.0` installed locally.
- A **Cloud-level** Confluent Cloud API key (not cluster-scoped) with permission to create environments, Kafka clusters, Flink compute pools, service accounts, role bindings, Flink artifacts, and statements. Generate via Console → Settings → Cloud API keys.

2.0 Provision + deploy the reports

```
make cc-flink-reports-up CONFLUENT_API_KEY=$CONFLUENT_API_KEY
CONFLUENT_API_SECRET=$CONFLUENT_API_SECRET
```

This runs scripts/deploy-cc-flink-reports.sh `create`, which builds the PTF shadow JAR if missing, then runs `terraform apply -auto-approve` in terraform/.

- **First run takes ~6–8 minutes** (Kafka cluster provisioning dominates).
- **Re-applies are idempotent** — `CREATE … IF NOT EXISTS` plus `lifecycle { ignore_changes = [compute_pool] }` on every statement.
- Optional fan-in provenance: `make cc-flink-reports-up ENABLE_MERGE_PROVENANCE=true …` (script flag `--enable-merge-provenance=true`, default `false`). See docs/flink-collector.md §2.4.
- Optional state-level provenance: `make cc-flink-reports-up ENABLE_STATE_PROVENANCE=true …` (script flag `--enable-state-provenance=true`, default `false`). Same version IDs as CP. See docs/state-provenance.md.
- Optional retention override: the script accepts `--day-count=<DAYS>` (default `30`); both `make` targets require `CONFLUENT_API_KEY` / `CONFLUENT_API_SECRET` or they fail fast with a clear message.

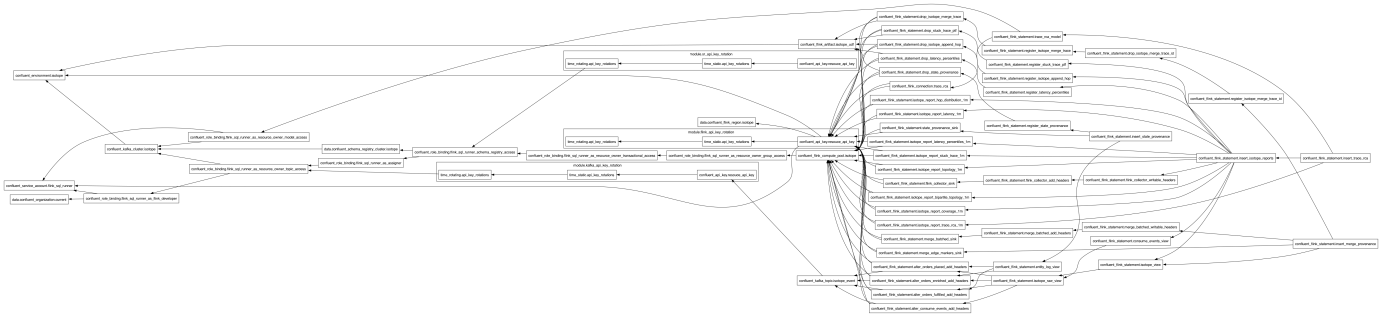
2.1 Verify in-band propagation

```
scripts/cc-app-run.sh place hello        # one traced record → orders.placed
scripts/cc-app-run.sh verify-inband      # asserts the Flink collector appended a hop
```

`verify-inband` reads a bounded snapshot of `orders.placed` and `orders.flink_enriched` and asserts that every trace on both topics kept its identity and gained **exactly one** hop, written by `flink-enrich`. Same trace ID with an incremented hop count is the proof that `ISOTOPE_APPEND_HOP` rehydrated the inbound isotope from the record's own headers rather than minting a fresh trace — the one propagation model that survives a Flink shuffle. It exits non-zero on failure, so it is usable as a CI gate. Pass a larger sample (`verify-inband 200`) if the two topics have drifted far apart.

The collector is a 1:1 projection with no window, so its output appears within seconds — the ~90s watermark wait applies only to the `TUMBLE` reports.

3.0 What gets created



Resource	Name	Notes
`confluent_environment`	`confluent-kafka-isotope`	ESSENTIALS stream-governance package
`confluent_kafka_cluster`	`kafka-isotope`	Standard, single-zone, AWS us-east-1 by default
`confluent_kafka_topic` × 4	`orders.{placed,enriched,fulfilled}` + `isotope_consume_edge_markers`	Only the event + consume-marker topics are pre-created. The 7 `isotope_report_*_1m` sink topics are created on first deploy by their `CREATE TABLE` (pre-creating them would make CCAF's Topic Catalog auto-import them as `(key BYTES, val BYTES)` and silently no-op the typed DDL).
`confluent_service_account` + 6 role bindings	`isotope-flink-sql-runner`	FlinkDeveloper (org) + ResourceOwner on topic/transactional-id/group/SR-subject `*` + Assigner
`confluent_flink_compute_pool`	`isotope-flink-statement-runner`	10 CFU; headroom for the 8-INSERT statement set + ad-hoc SELECTs
`confluent_flink_artifact`	`isotope-flink-udf`	Uploads `ptf/build/libs/isotope-flink-udf.jar`
`confluent_flink_statement` × 28	(see file)	6 ALTER TABLE + 3 VIEW + 8 sink CREATE TABLE + 5 CREATE FUNCTION (2 PTFs + 3 UDFs) + 1 EXECUTE STATEMENT SET holding all 8 INSERT INTOs (23 long-lived) + 5 transient DROP FUNCTION. The 2 merge UDFs are registered even when `var.enable_merge_provenance` is `false` — registration is inert until a statement calls it.
`confluent_flink_statement` × 3 (optional)	(see terraform/setup-ccaf-ai.tf)	Optional AI trace-RCA report — `CREATE MODEL trace_rca` + 1 Protobuf sink + 1 `INSERT … ML_PREDICT`. **Gated on `var.enable_trace_rca` (default `false`)**, so a normal apply skips them entirely. Set

Resource	Name	Notes
		`rca_model_api_key` (and `rca_model_provider`/`_version`/`_endpoint` for a non-OpenAI provider) to enable. See root README §3.3.
`confluent_flink_statement` × 5 (optional)	(see terraform/setup-ccaf-state-provenance.tf)	Optional state-level provenance — sink `CREATE TABLE` for `isotope_state_provenance` (append mode, `'kafka.retention.time' = '0'` for unlimited retention), the `entity_log` view, `DROP` + `CREATE FUNCTION STATE_PROVENANCE`, and its own `INSERT`. **All 5 gated on `var.enable_state_provenance` (default `false`)**: the function registration is gated too, because CCAF checks a PTF's state type at `CREATE FUNCTION` time. Version IDs match CP's. See docs/state-provenance.md.
`confluent_flink_statement` × 5 (optional)	(see terraform/setup-ccaf-merge-provenance.tf)	Optional fan-in (merge) provenance — 2 sink `CREATE TABLE` + 2 header `ALTER TABLE` + 1 statement set holding the merge INSERT and its merge-edge INSERT. **Gated on `var.enable_merge_provenance` (default `false`)**. The 4 `CREATE`/`DROP FUNCTION` statements registering `ISOTOPE_MERGE_TRACE` / `ISOTOPE_MERGE_TRACE_ID` are *not* gated — registration is inert until a statement calls it. See docs/flink-collector.md §2.4.

Two rotating service-account API key pairs (one Kafka, one Schema Registry) are managed by `module.kafka_api_key_rotation` and `module.sr_api_key_rotation` in terraform/setup-cc-kafka.tf.

4.0 Useful outputs

```
terraform -chdir=terraform output environment_id
terraform -chdir=terraform output kafka_bootstrap_servers
terraform -chdir=terraform output schema_registry_url
terraform -chdir=terraform output -raw kafka_api_key      # sensitive
terraform -chdir=terraform output -raw kafka_api_secret   # sensitive
```

5.0 Drive traffic (required to see report rows)

The demo app runs **on your host** and talks to Confluent Cloud over SASL_SSL. You don't export credentials manually: scripts/cc-app-run.shsources scripts/cc-cli-env.sh, which pulls the Kafka + Schema-Registry credentials from `terraform output`, builds the JAAS string, and invokes `./gradlew :app:run` with the six `-D` flags. It hard-fails if any of the seven required values is missing.

Run the 4-stage demo, one verb per terminal (same A/B/C/D order as the local demo). B/C/D are long-running (Ctrl-C to stop). Start order doesn't matter: each stage reads its topic from the earliest offset with a fresh consumer group, so it replays everything already there. Starting B/C/D first just lets you watch records flow live.

```
scripts/cc-app-run.sh place 'hello'    # A — kick the chain off (orders.placed)
scripts/cc-app-run.sh enrich           # B — orders.placed   → orders.enriched
scripts/cc-app-run.sh fulfill          # C — orders.enriched → orders.fulfilled
scripts/cc-app-run.sh ship             # D — terminal consume orders.fulfilled (emits
marker)
```

To skip the backlog, pass `--latest` (shorthand for `-Disotope.consume.from=latest`), e.g. `scripts/cc-app-run.sh --latest enrich`. Each stage then sees only records produced after it starts, so you **must** start B/C/D before A. It's also the mode to use when reading the latency report: a replayed backlog record's origin→hop latency is `now - origin_ts`, which dwarfs steady-state latency.

`cc-app-run.sh` also accepts the generic `send` / `hop` / `consume` / `sink` passthrough for ad-hoc topics — run it with no args for the full verb list.

Sustained traffic. The report jobs aggregate over `TUMBLE(event_time, INTERVAL '1' MINUTE)` windows, which only emit once the watermark advances past `window_end`. A burst inside a single 1-minute interval sits in one open window forever — spread records across **multiple** windows:

```
# 30 records, 5s apart ≈ 2.5 min of event-time → spans 3+ windows
for i in {1..30}; do scripts/cc-app-run.sh place "burst-$i"; sleep 5; done
```

Wait ~90s after the **last** record before checking results — that's the watermark catching up. `stuck_trace_alerts_1m` only fires for a trace that goes ≥60s of event time without a fresh hop and has not reached `orders.fulfilled`; to exercise it, send one record to `orders.placed`, skip the `enrich`/`fulfill` hops, and keep sending unrelated records so the watermark crosses `event_time + 60s`.

6.0 Observe

In the Cloud Console **Flink SQL workspace**, query the report tables:

```
SELECT * FROM isotope_report_latency_1m;
SELECT * FROM isotope_report_bipartite_topology_1m;   -- all 6 edges: 3 produce + 3
consume
SELECT * FROM isotope_report_hop_distribution_1m;
-- …coverage, stuck_trace, topology, latency_percentiles
```

Terminal D prints the same `trace_id` across all three hops. CCAF report topics ride **Protobuf+SR** (`proto-registry`), versus Avro+SR on the CP path.

With `ENABLE_MERGE_PROVENANCE=true`, you can see which merged records a stuck trace fed. Paste the two queries from `82_merge_impact_query.fql` into the workspace, replacing `stuck_trace_alerts_1m` with `isotope_report_stuck_trace_1m`. Both run until you stop them, so stop them once you have your answer.

7.0 Teardown

```
make cc-flink-reports-down CONFLUENT_API_KEY=$CONFLUENT_API_KEY
CONFLUENT_API_SECRET=$CONFLUENT_API_SECRET
```

Runs `terraform destroy -auto-approve` — deletes **every** resource above, including the environment itself (the report sink topics are cleaned up via the environment cascade). Safe to run repeatedly.

> **Cost note:** the Kafka cluster and compute pool bill while they exist. Teardown when you're done rather than leaving the environment running.

8.0 Troubleshooting

- **`CONFLUENT_API_KEY … must be set`.** The `make` target requires both vars on the command line (or exported and passed through) — see §1.0 Prerequisites.
- **`Aggregate functions are not supported`.** Expected if you try to add a UDAF — CCAF rejects user-defined aggregates, which is why percentiles is a PTF — see root README §3.3.
- **Typed `CREATE TABLE` silently no-ops.** A report sink topic was pre-created, so the Topic Catalog auto-imported it as `(key BYTES, val BYTES)` — don't pre-create the `isotope_report_*_1m` topics — see §3.0 What gets created.
- **No report rows.** Almost always the watermark — traffic must span multiple 1-minute windows; wait ~90s after the last record — see §5.0 Drive traffic.
- **`Column headers already exists in the table.`** The `ALTER TABLE … ADD (headers …)` statements are not idempotent and CCAF has no per-column `IF NOT EXISTS`; the column belongs to the topic's catalog entry and outlives the statement resource, so a re-created statement fails and leaves the resource tainted, blocking everything downstream. `make cc-flink-reports-up` now untaints them automatically before applying, including the count-gated `merge_batched_add_headers[0]`. Running terraform by hand? Do it yourself first: `terraform state list | grep -E '_add_headers(\[[0-9]+\])?$' | xargs -n1 terraform untaint`. The usual trigger was a Flink API key rotation. It changes every statement's `credentials`, and the deploy script used to turn that into a forced replacement of every statement. It now applies a credentials-only change in place, which the provider has supported since 2.84.0, so a rotation re-creates nothing (#448).
- **`error reading Flink Statement … 401 Unauthorized`.** Terraform reads a statement back with the API key recorded in *state*, and `module.flink_api_key_rotation` keeps only `number_of_api_keys_to_retain` keys — so a statement that outlives that many rotations points at a deleted key, and refresh fails on **both** apply and destroy (the environment becomes un-destroyable). `make cc-flink-reports-up`/`-down` now runs `scripts/tf-repair-flink-credentials.py` first, which re-points those statements at a key that still exists. Run it by hand if you drive terraform yourself: `python3 scripts/tf-repair-flink-credentials.py terraform/terraform.tfstate`.
- **`Request is malformed. name needs to contain lowercase alphanumeric characters and hyphens only`.** A `statement_name` used underscores. Statement names are Kubernetes-style (`isotope-report-latency-1m`); the sink *table* names keep their underscores (`isotope_report_latency_1m`) — the two are not the same string.
- **`"stopped" or "properties_sensitive" attribute must be updated for Flink Statement`.** A statement resource changed in a way the provider cannot apply in place — edited SQL, a new `statement_name`, different properties. Terraform plans it as an in-place update and the API rejects it; it has to be a destroy-and-recreate. `make cc-flink-reports-up` detects these from the plan and re-runs them as `-replace=` automatically; by hand, pass `-replace=confluent_flink_statement.<name>` to `terraform apply`.
- **App can't authenticate.** `cc-cli-env.sh` couldn't read `terraform output` — re-run `make cc-flink-reports-up` so the rotated keys exist, then retry.